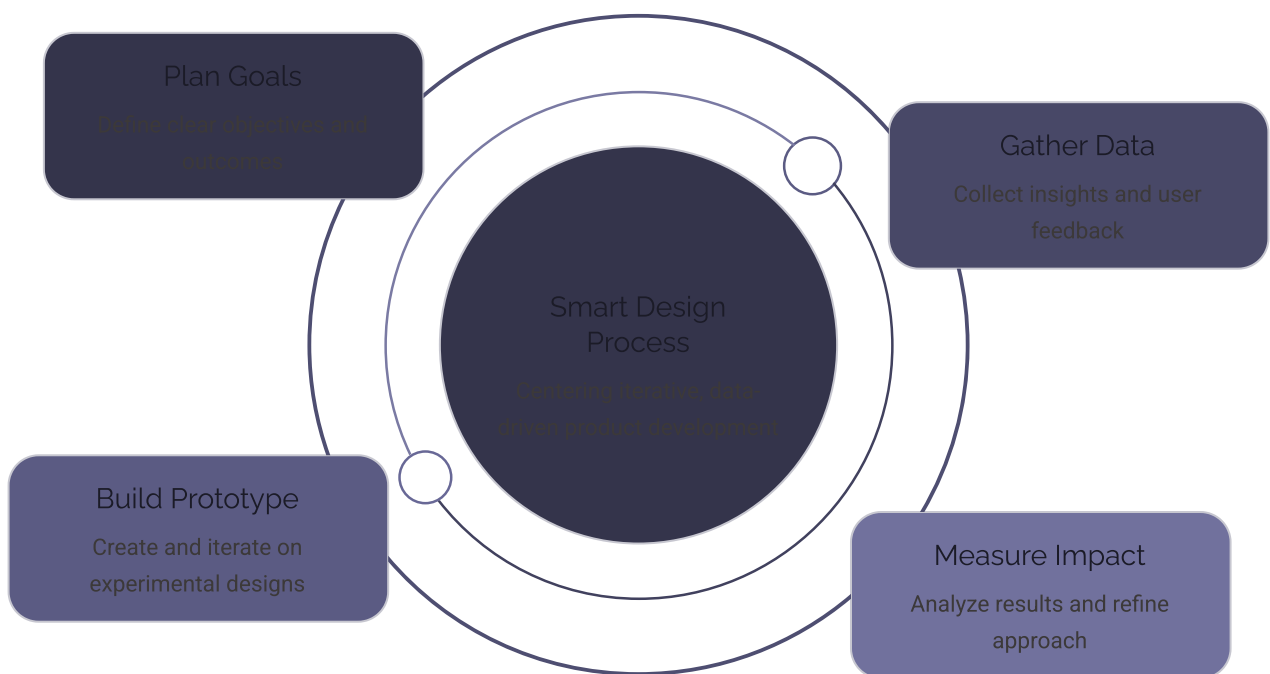
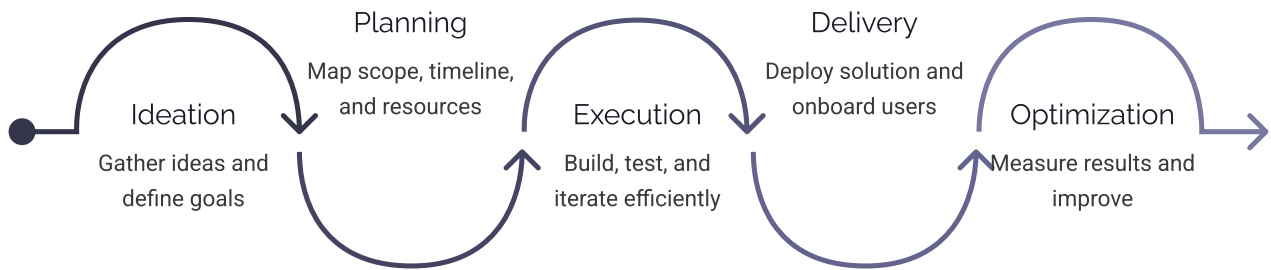






Assessment of Coral Reef Health: Equipment, Survey Methods, and Practical Criteria for Distinguishing Live and Dead Coral

Why Coral Reefs Matter

<0.1% of Ocean Floor

Yet support approximately **25% of all marine species** — the highest biodiversity per unit area of any marine ecosystem.

Coastal & Fishery Services

Protect coastlines from storm surge and sustain global fisheries that millions depend on for food security.

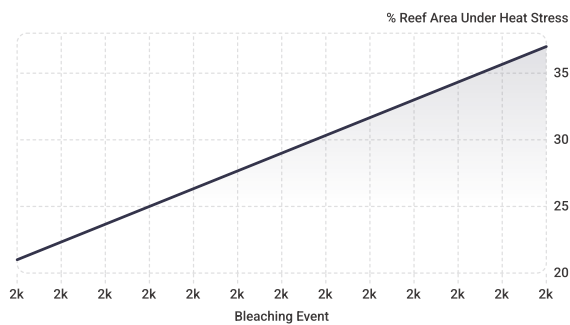
Symbiotic Foundation

Built on the partnership between stony corals and *Symbiodiniaceae* algae — a bond vulnerable to thermal stress.

The Bleaching Mechanism

Rising ocean temperatures collapse this symbiosis — algae are expelled, coral tissue turns white and begins to starve.

The Escalating Crisis: GCBE4



Unprecedented Scale

The Fourth Global Bleaching Event (GCBE4) is the **most widespread and severe on record**, affecting reef systems across 83+ countries.

From 21% in 1998 to 84.4% in 2023–2025 — a four-fold increase in the share of global reef area exposed to thermal stress.

 NOAA was forced to expand its alert system to capture the unprecedented scale of GCBE4.

Research Objectives

1	Standardize Equipment Define and validate gear requirements for both shallow and deep-water reef survey contexts.
	Establish Diagnostic Criteria

- | | |
|---|--|
| 2 | Operationalize clear, replicable classification of live, dead, partially dead, and bleached coral tissue. |
| 3 | <p>Integrate Satellite Monitoring</p> <p>Connect NOAA Degree Heating Weeks data with field-based metrics for cross-scale validation.</p> |
| 4 | <p>Propose Coral Mortality Index</p> <p>Deliver a single, quantitative framework that benchmarks reef trajectory across sites and over time.</p> |

Four Core Survey Methods

Line Intercept Transects

Measuring tape deployed along a depth contour; substrate changes recorded to yield **linear** % **cover** estimates across the reef.

Quadrat Sampling

Fixed frames (0.5–1 m²) placed at intervals to capture fine-scale spatial variability within a defined area.

Photo-transect Photogrammetry

Digital overlays with 20–50 random points per image reduce observer bias and minimize required bottom time.

Visual Bleaching Assessment

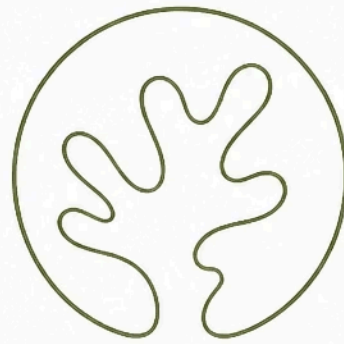
CoralWatch Health Chart: scored 1 (fully bleached) to 6 (fully pigmented), enabling standardized field comparisons.

Diagnostic Criteria: Live vs. Dead



Live Coral

(intact pigmented tissue,
active polyp response)



Bleached Coral

(intact but translucent tissue,
polyps under severe thermal stress)

Handling Field Complications

Partial Mortality

When a single colony is part-alive and part-dead, binary classification fails. Score the dead portion as a **percentage of colony area** — this preserves quantitative resolution and feeds directly into the Coral Mortality Index.

Active Disease Lesions

Black Band Disease, White Syndrome, and Stony Coral Tissue Loss Disease (SCTLD) each present with **active tissue-loss margins moving across the colony in real time**. Record lesion position, direction of advance, and estimated progression rate — not just presence/absence.

The Coral Mortality Index

CMI Formula

$$\text{CMI} = \frac{\text{Dead Coral Cover}}{\text{Live} + \text{Dead Coral Cover}} \times 100$$

A single, dimensionless percentage that captures reef condition at any point in time and remains directly comparable across sites and surveys.

Interpretation Thresholds

CMI < 20%

Low-mortality state – reef system largely intact

CMI 20–50%

Moderate impact – trajectory monitoring essential

CMI > 50%

Severe degradation – intervention and escalated monitoring warranted

Bridging Field Data & Global Satellite Monitoring



NOAA Coral Reef Watch

Provides 5 km-resolution satellite data on sea surface temperature and **Degree Heating Weeks (DHW)** – the primary global indicator of thermal bleaching risk.

A Bidirectional Validation Loop

Field bleaching scores ground-truth satellite models; satellite data in turn guides survey prioritization. Neither stream is complete without the other.

- Satellite data identifies **where to survey**
- Field data **improves satellite accuracy** and calibrates predictive models

Safety, Ethics & Compliance



Permits & MPA Compliance

Research permits and marine protected area compliance must be secured **before any fieldwork begins**. No exceptions.



Diver Qualifications

All field personnel must hold current diver certification and valid medical clearance appropriate to survey depth.



Low-Impact Protocols

Neutral buoyancy at all times, reef-safe sunscreen only, and full equipment disinfection between sites to prevent cross-contamination.

Integrated Framework for Global Reef Monitoring

Standardized equipment, replicable field methods, and clear diagnostic criteria — fully operationalized into a deployable protocol.



Standardized Field Methods

Four replicable survey techniques with defined equipment and diagnostic criteria

Coral Mortality Index

2

Quantitative, comparable metric that tracks reef trajectory across sites and time

3

Global–Local Integration

84% of the world's reefs under heat stress — this framework connects local observation to the global picture